as we'll do next. The marker we define can be any sort of SVG shape, but we'll use a path because it lets us draw an arrowhead. A marker can be drawn at the start, end, or middle of a line, and has settings to determine its direction relative to its parent element.

Listing 6.8 Marker definition and application

```
var marker = d3.select("svg").append('defs')
      .append('marker')
      .attr("id", "Triangle")
      .attr("refX", 12)
      .attr("refY", 6)
      .attr("markerUnits", 'userSpaceOnUse')
      .attr("markerWidth", 12)
      .attr("markerHeight", 18)
      .attr("orient", 'auto')
      .append('path')
      .attr("d", 'M 0 0 12 6 0 12 3 6');

d3.selectAll("line").attr("marker-end", "url(#Triangle)");
```

The default setting for markers bases their size off the stroke-width of the parent, which in our case would result in difficult-to-read markers.

A marker is assigned to a line by setting the marker-end, marker-start, or marker-mid attribute to point to the marker.

With the markers defined in listing 6.9, you can now read the network (as shown in figure 6.9) more effectively. You see how the nodes are connected to each other, and you can spot which nodes have reciprocal ties with each other (where nodes are connected in both directions). Reciprocation is important to identify, because there's a big difference between people who favorite Katy Perry's tweets and people whose tweets are favorited by Katy Perry (the current Twitter user with the most followers). Direction of edges is important, but you can represent direction in other ways, such as using curved edges or edges that grow fatter on one end than the other. To do something like that, you'd need to use a `<path>` rather than a `<line>` for the edges like we did with the Sankey layout or the arc diagram.

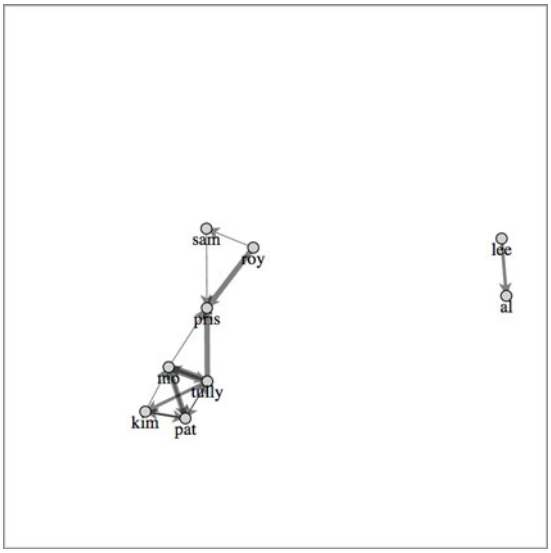


Figure 6.9 Edges now display markers (arrowheads) indicating the direction of connection. Notice that all the arrowheads are the same size.